

Shubham Jain



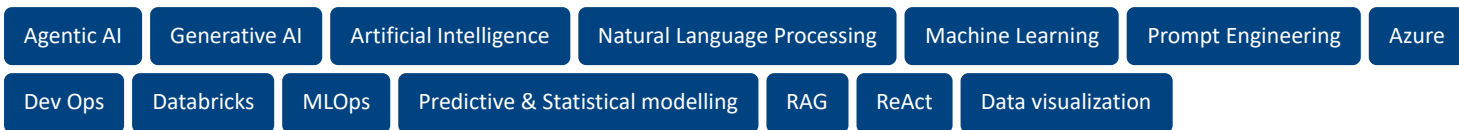
Senior AI Scientist (Azure certified AI Engineer)

+91-9043874159 jainshubh2207@gmail.com Bangalore <https://www.linkedin.com/in/shubham-jain-2b38089a/>

SUMMARY

Innovative and impact-driven AI professional with 11 years of experience in designing and deploying advanced AI/ML/GenAI solutions across Healthcare, Banking, Retail, and Manufacturing. Proven ability to lead multidisciplinary teams, architect scalable AI platforms, and translate business needs into actionable AI strategies. Adept at working with LLMs, RAG, multi-agent systems, and cloud-native tools to deliver high-impact enterprise solutions that reduce costs and drive efficiency.

TECHNICAL SKILLS



PROFESSIONAL EXPERIENCE

Senior AI Scientist

Apr '20 - Present

AI Labs at Course5 Intelligence Pvt. Ltd.

Bangalore, IN

Leading AI initiatives across Fortune 500 clients, with end-to-end ownership of design, architecture, deployment, and optimization of GenAI and ML solutions.

- Streamlined enterprise **information retrieval** for a global **FMCG** client by deploying a **multi-agent chatbot** on **Azure Databricks**, architected with **LangGraph** and **ReAct**-based **RAG** framework.
- Established a complete **LLMOps** lifecycle using **MLflow** for **production tracing**, **prompt registry**, **capture user feedback**, and “LLM-as-a-judge” **scoring** all operationalized via **CI/CD** pipeline.
- Deployed an end-to-end **GenAI** analytics platform on Azure for a **Fortune 500 healthcare** client, leveraging LangChain RAG architecture with **BERT** for intent recognition to handle complex Text-to-SQL queries, all automated via CI/CD and **ADF** pipelines to **cut manual reporting by 80%**.
- Implemented advanced Text-to-SQL capabilities on **Databricks Genie**, strengthening **data governance** and **security** with user-level access control via **Unity Catalog**.
- Fine-tuned** a **Mistral 7B** model for a text-to-SQL task, leveraging **QLORA PEFT** technique and 8/4-bit **quantization** to reduce memory footprint and lower inference latency.
- Developed a **knowledge management** solution employing **Azure Cognitive Services** and **azure search** to automate metadata tagging and document summarization, resulting in a 70% reduction in manual effort.
- Led the pre-sales process for GenAI and ML solutions**, designing complex agentic AI architectures and developing impactful proofs-of-concept with detailed API and cost modeling to secure key enterprise accounts.
- Collaborated with cross-functional stakeholders** (product, engineering, business) to integrate AI capabilities into enterprise workflows.
- Mentoring a 5-member AI team** while leading **Agile** ceremonies to accelerate development velocity and establishing best practices in AI/ML development and delivery.

Senior Data Scientist

Oct '18 - Mar '20

Tata Consultancy Services

Bangalore, IN

Delivered predictive ML solutions for clients in banking and manufacturing, with a focus on scalable deployment and measurable business impact.

- Boosted cross-sell conversion by 20% for a major **bank** by engineering an automated **ML pipeline** that used **Logistic Regression** and **Random Forest** models for batch inference and **retraining**, delivering actionable churn predictions and sales leads directly to a performance **dashboard**.
- Built an ML-powered **claims settlement engine** for a paper board manufacturing client using a predictive **scoring model** with **Logistic Regression** and **Weight of Evidence (WOE)** technique, which automated 82% of submissions and slashed processing time from 25 days to one.
- Ensured **model transparency** by analyzing model coefficients and applying post-hoc techniques like **SHAP**, **LIME**, and **Feature Importance** for local and global **model explanations**.

Data Scientist

Aug '17 - Oct '18

Mercedes Benz R&D India

Bangalore,IN

Pioneered analytics solutions for automotive warranty, diagnostics, and reporting processes.

- Improved warranty cost forecasting by developing ML models (**SVR**, **Random Forest**) and deploying their outputs into a dynamic **dashboard**, empowering the finance department to more confidently allocate reserves and anticipate liabilities based on Repairs Per Hundred Engines (RPH) and Cost Per Engine (CPE) KPIs.
- Automated reporting pipeline using **Python** and rule-based **NLG** to translate **SQL** data into weekly KPI presentations, slashing manual effort by 90%.
- Executed a rigorous model selection process by benchmarking multiple algorithms through k-fold **cross-validation** and **hyperparameter** tuning to identify the model with the lowest prediction error (**RMSE/MAE**) on business KPIs.

Software Developer

Aug '14 - Aug '17

Robert Bosch India

Coimbatore,IN

Delivered IoT and analytics-backed solutions for automotive use cases, with emphasis on cost and process optimization.

- Developed a hybrid **predictive maintenance** engine for BEV powertrains, integrating **statistical analysis**, rule-based logic, and **anomaly detection** models to cut operational costs by \$5M through proactive fault detection.
- Designed **data visualizations** in **Tableau** to enhance analytical storytelling and decision-making.
- Automated **data extraction and cleansing pipelines**, while ensuring robust delivery via **unit and integration testing**.

Tools

LLMs: GPT4o , Sonnet, Gemini

Frameworks: Pytorch, Tensorflow, Keras, MLflow

Databases: SQL, MongoDB, CosmosDB, Neo4J

Languages: Python, R

Operating system: Linux, Windows

Libraries: Langchain, HuggingFace, LangGraph, Spacy

EDUCATION

B.Tech - Electrical & Electronics Engineering

Jul '10 - May '14

NIT Trichy

Tiruchirappalli, TN

Certifications

- Microsoft certified Azure AI Engineer Associate | Jan 2025
- Neo4j Certified Professional | Mar 2022
- Coursera Deep learning Specialization | Mar 2019
- Coursera Machine Learning Course by Andrew Ng | May 2017